

SKY BREEZE ACADEMY

Group of Sky Breeze AC Solutions

Air Conditioning & HVAC Training Institute

Coimbatore, Tamilnadu

Course Title : Industry-Oriented Internship Program in HVAC & Refrigeration Engineering

Program Type : 3-Month Full-Time Technical Internship (Office + Site Based)

(Engineering-Focused)

+91 94860 77714 | +91 94861 58877

Eligible Branches:

- Mechanical Engineering
- Electronics & Instrumentation Engineering (E&I)
- Electrical & Electronics Engineering (EEE)
- Production Engineering

COURSE OBJECTIVE

To bridge the gap between engineering theory and real-world HVAC systems, enabling students to understand HVAC from a system design, site execution, and industry perspective, and prepare them for HVAC-related engineering roles.

SYLLABUS STRUCTURE

MODULE 1: FUNDAMENTALS OF HVAC ENGINEERING

- Overview of HVAC systems and applications
- Thermal comfort parameters and standards
- Psychrometric properties of air
- Sensible and latent heat concepts
- Introduction to HVAC load components

MODULE 2: REFRIGERATION PRINCIPLES

- Vapour compression refrigeration cycle
- Refrigerants and their properties
- Compressor working principles
- Condensers, evaporators, and expansion devices
- System performance parameters (COP, efficiency)

MODULE 3: HVAC SYSTEM CLASSIFICATION

- Room Air Conditioning (RAC / MRAC systems)
- Light commercial air conditioning systems
- Centralized HVAC systems
- Application-based system selection
- Comparison of DX and chilled water systems

MODULE 4: VRF & VARIABLE CAPACITY SYSTEMS

- VRF / VRV system architecture
- Indoor and outdoor unit configurations
- Inverter-driven compressor operation
- Refrigerant piping concepts
- Operational and maintenance considerations

MODULE 5: AIR DISTRIBUTION & TERMINAL UNITS

- Air handling units (AHU)
- Fan coil units (FCU)
- Ductable units and package units
- Air distribution components (ducts, diffusers, grilles)
- Zoning and airflow control concepts

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MODULE 6: CHILLER & CENTRAL PLANT SYSTEMS

- Chiller system overview and applications
- Air-cooled and water-cooled chillers
- Cooling towers and condenser water systems
- Vapour compression and absorption chillers
- Central plant operation fundamentals

MODULE 7: HVAC COMPONENTS & EQUIPMENT

- Compressor types and selection criteria
- Heat exchangers and coil construction
- Expansion valves and control devices
- Pumps, fans, and motors
- Filtration and indoor air quality components

MODULE 8: HVAC ELECTRICAL & CONTROLS

- Power consumption characteristics of HVAC systems
- Inverter technology and part-load operation
- Electrical components used in HVAC equipment
- Basic control logic and safety interlocks
- Introduction to BMS and automation systems

MODULE 9: INSTALLATION, COMMISSIONING & MAINTENANCE

- HVAC installation practices and standards
- Refrigerant piping and insulation basics
- System commissioning procedures
- Performance testing and balancing
- Maintenance philosophy and lifecycle considerations

MODULE 10: HVAC INDUSTRY PRACTICES & CAREER ROLES

- HVAC project lifecycle
- Design, site execution, and O&M roles
- Industry expectations from engineering graduates
- Skill requirements for HVAC engineers
- Career pathways in HVAC industry

 INTERNSHIP DELIVERY MODE

- Classroom-based technical sessions
- Office exposure to drawings and documentation
- Site visits and system observation
- Interaction with working HVAC engineers
- Short-term field assignments (as applicable)

 CERTIFICATION

- Internship Completion Certificate issued by **SkyBreeze Academy (Group of SkyBreeze AC Solutions)**

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